Neuronal phenotyping for drug discovery

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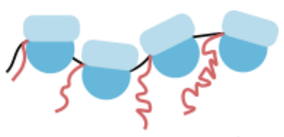
? → Can Parkinson's disease drug discovery be performed in a disease-relevant cell type using screening-suitable technology?

LRRK2 kinase overactivity triggers Parkinson's disease-related mechanisms

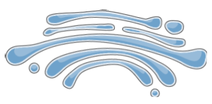
- Leucine-rich repeat kinase 2 (LRRK2) mutations account for roughly 1% of sporadic and 4% of familial PD cases
- LRRK2 kinase domain G2019S mutation leads to kinase overactivity

Facilitates misfolded protein buildup



Increased 4E-BP & 40S phosphorylation

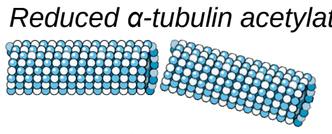
Compromises vesicular sorting



Increased RAB phosphorylation & inactivation

Decreases microtubule stability & neurite outgrowth

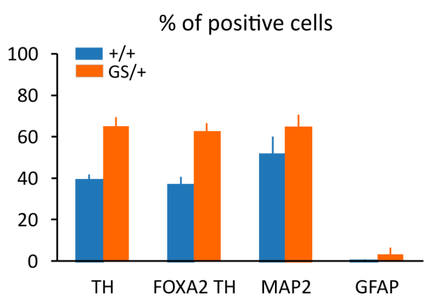
Reduced α-tubulin acetylation



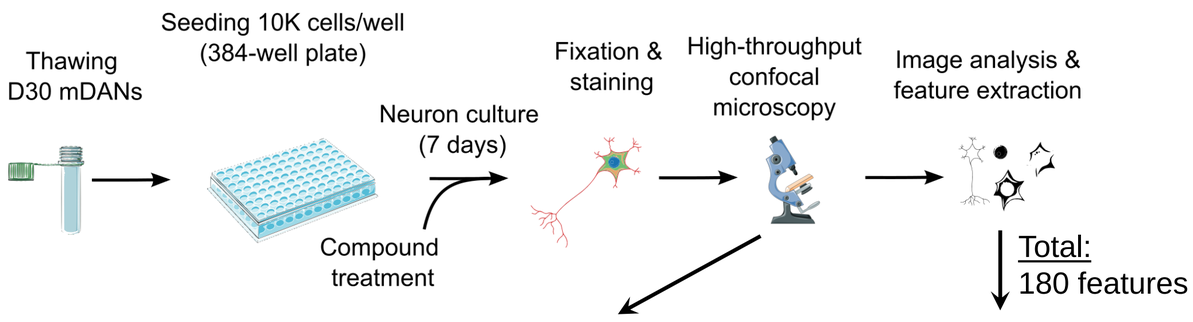
Impairs mitochondrial quality control

Altered mitophagy

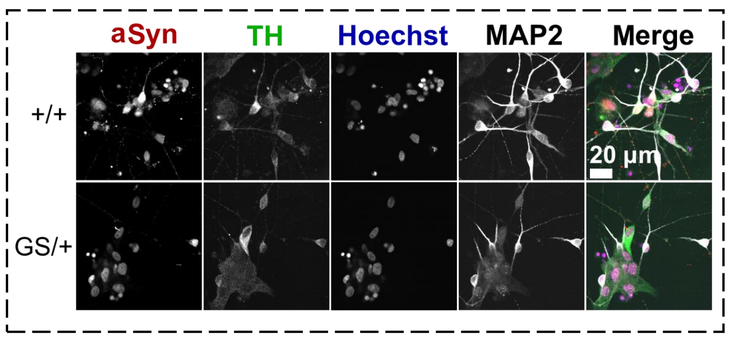
- **GS/+**: LRRK2 G2019S-patient iPSCs differentiated into midbrain dopaminergic neurons (mDANs)
- **+/+**: Mutation-corrected isogenic iPSCs differentiated into mDANs



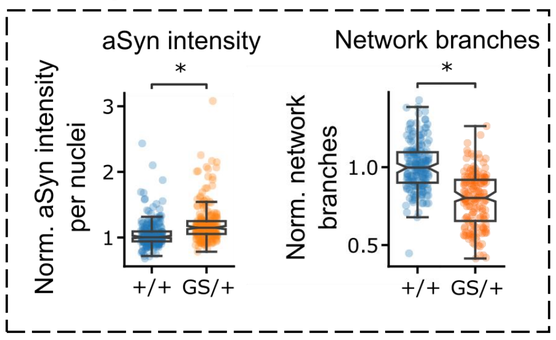
Parkinson's disease modelling in human patient iPSC-derived midbrain dopaminergic neurons (mDANs)



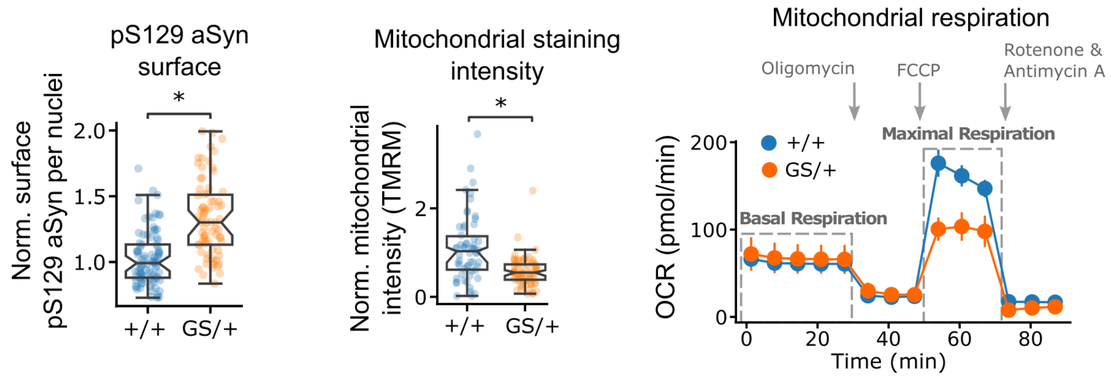
Staining panel in D37 isogenic mDAN pair



Example neuronal features

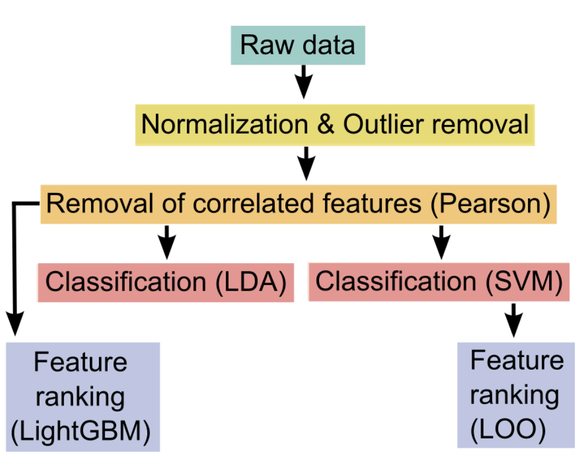


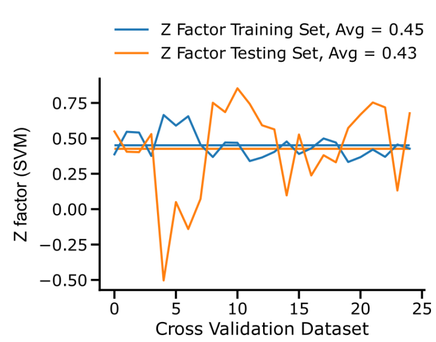
Additional hallmarks of Parkinson's disease in D37 isogenic mDAN pair

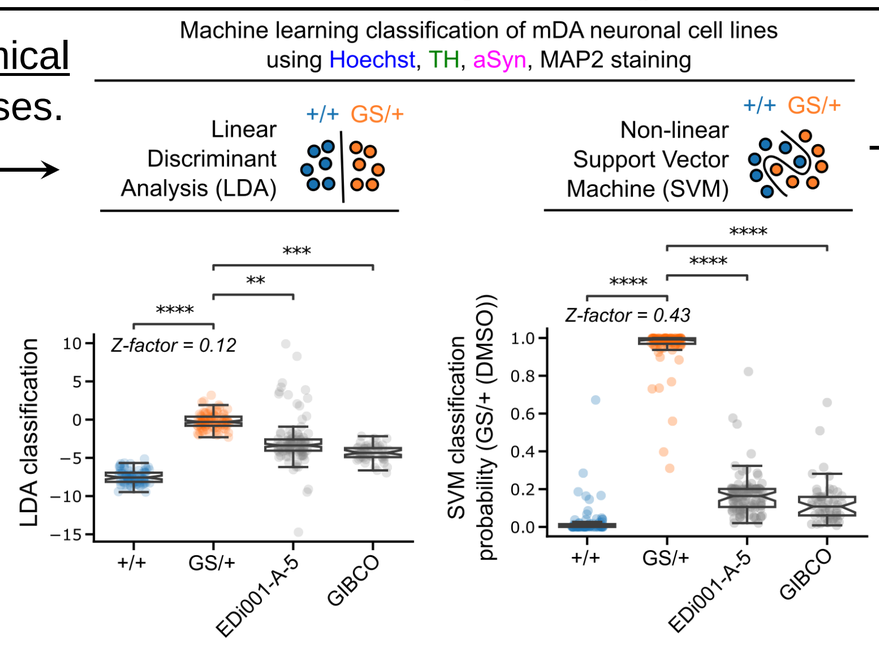


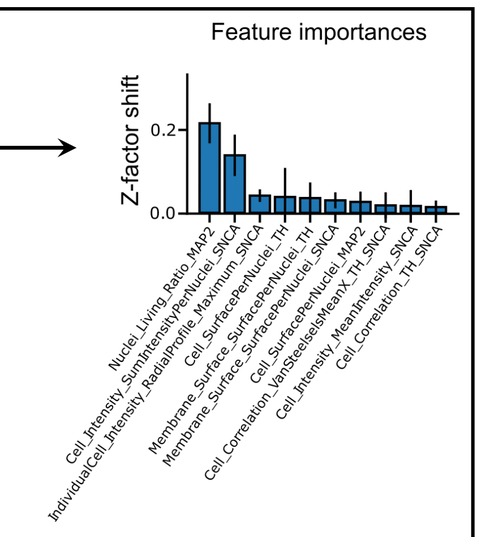
Machine Learning (ML) strategy to classify neurons based on multiple image-derived cellular features

"Unseen" conditions such as novel cell lines or tested chemical compounds can be mapped relative to the two reference classes.



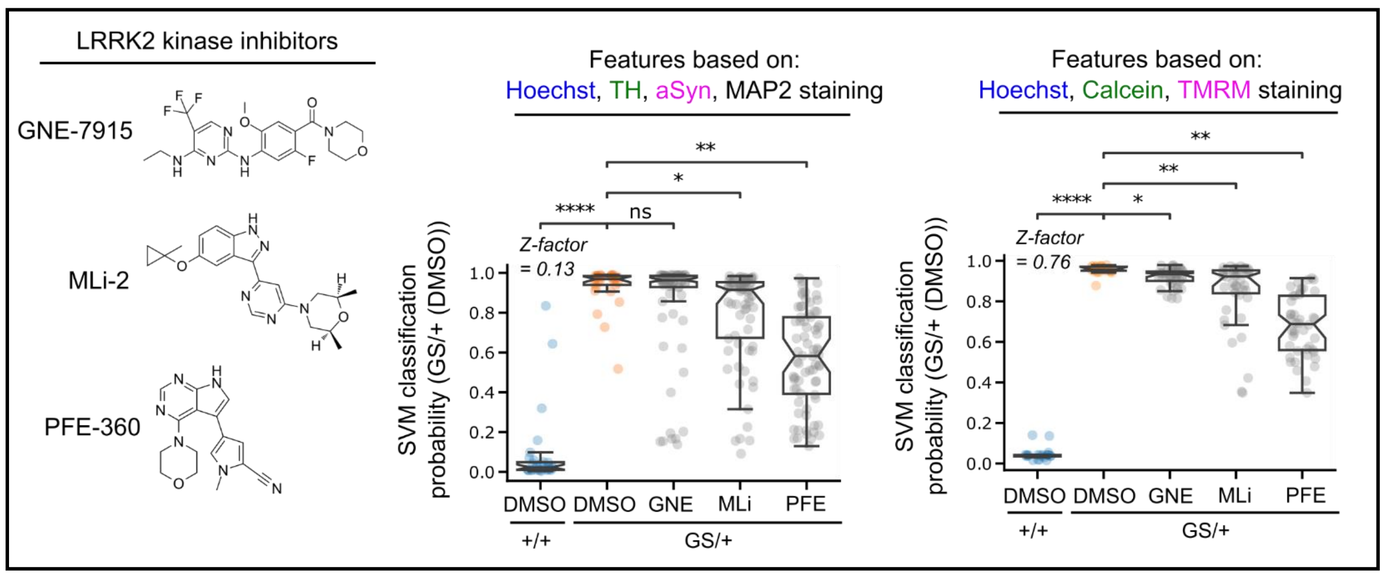




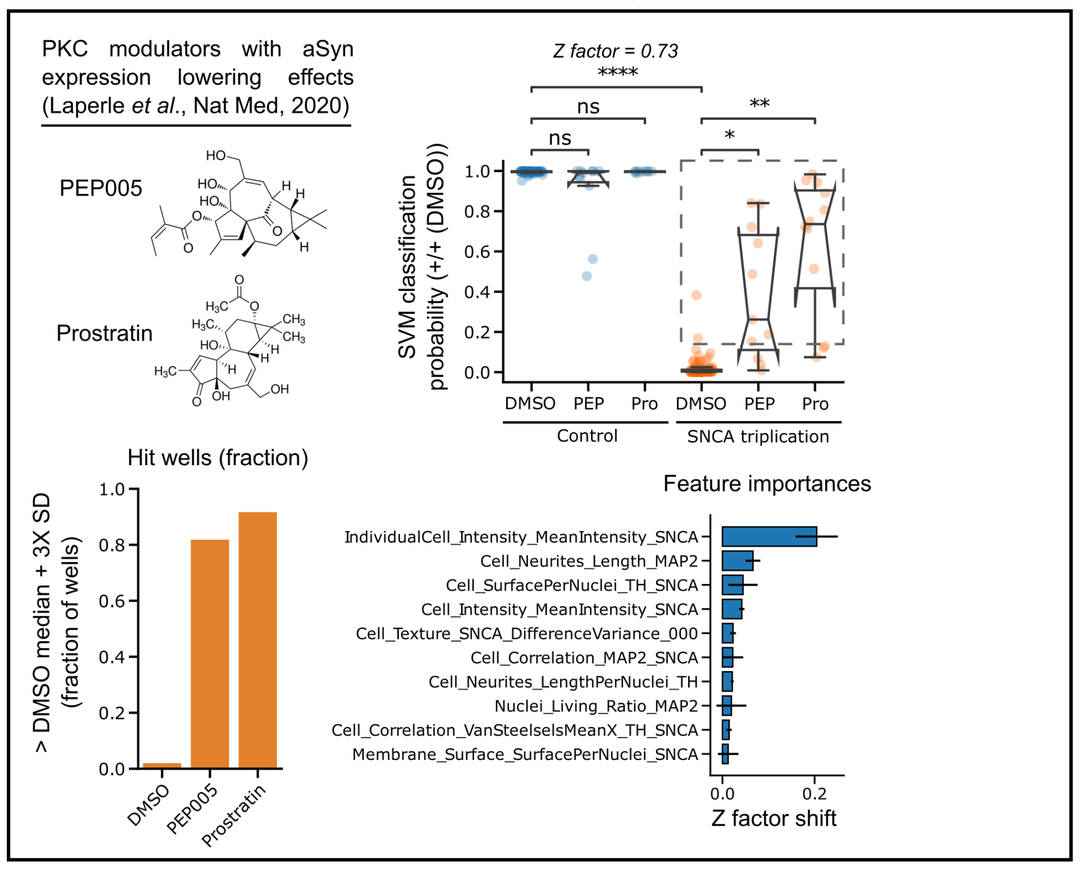


"Leave-one-out" (LOO) to determine feature importances.

Multi-feature detection of LRRK2 kinase inhibition effects



Multi-feature detection of aSyn reduction effects



Highlights

- Parkinson's disease phenotypes are present in two midbrain neuronal models
- Assay automation and adaption to 384-well plates
- Machine learning classifies neuronal genotypes based on image-extracted features
- Machine learning identifies drug treated neurons based on image-extracted features